

Artificial intelligence - partner relationships management for climate management in B2B firms to achieve sustainable competitiveness

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ABSTRACT

Drawing on the dynamic capability view and resource-based view, this study investigates the relationship among artificial intelligence (AI)-based partner relationship management, firm performance, and sustainable competitiveness concerning the climate management of business-to-business (B2B) firms. Throughout this investigation, the study additionally explored the impact of several other dimensions, including information communication technology capability, firm fit, and technological readiness, as fundamental requirements for the implementation of AI-based partner relationship management. The present study also investigates the relationship between firm partner engagement and firm information processing systems for climate management of B2B firms and their impact on achieving firm performance and sustainable competitiveness. A mixed-methods approach was employed that involves in-depth interviews with senior managers from a diverse set of B2B firms, and the proposed hypothesis model was evaluated by analysing the collected B2B data of 142 responses using partial least squares structural equation modelling. The findings of the study show that information communication technology capability and technological readiness play a significant role in improving the performance of AI-based partner relationship management. Furthermore, the implementation of AI-based partner relationship management in B2B firms would help partner engagement and information processing systems for climate management, leading to sustainable firm competitiveness. The outcomes of this study offer many implications for managers and practitioners in the B2B sector and suggest potential avenues for future research.

1. Introduction

Firms are beginning to see the value of combating climate change, which is a major global concern. The United Nations Climate Change “Conference of the Parties” is an annual event where global leaders, policymakers, experts, and stakeholders come together to debate and negotiate solutions to climate change¹. The conference's goal is to facilitate international collaboration on climate change prevention and adaptation, and it serves as a stage for governments to present their progress in these areas.¹ As the effects of climate change become more widely understood, firms in all sectors have begun to put more emphasis

on environmentally responsible operations (Coppola & Blohmke, 2019). Firms that cater to other businesses have the potential to make an enormous difference by taking action on climate change. Taking measures to combat climate change has several potential benefits for business operations. First, it may aid in cost reduction by enhancing energy efficiency, reducing waste, and decreasing carbon emissions (Water, 2023). Additionally, it may improve a firm's image and give it an advantage in a crowded market, both of which are crucial for business-to-business (B2B) firms. Furthermore, combating climate change may inspire the creation of cutting-edge technologies and novel commercial offerings, generating additional sources of income. By placing a

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¹ <https://unfccc.int/climate-action/introduction-climate-action> (accessed on 19/04/2023)

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premium on addressing climate change, B2B firms may earn investors' trust and gain easier access to funding (Layne, 2023). In recent years, several B2B firms have been exemplary in this area in that they have taken action to combat climate change and have experienced positive results. For example, Unilever has made substantial efforts in terms of renewable energy, energy effectiveness, and sustainable packaging, contributing to its decreased carbon footprint and enhanced consumer reputation.² Microsoft's investments in renewable energy, energy efficiency, and carbon removal technology have helped the corporation lower its carbon footprint and enhance its image, which has contributed to the company functioning better financially than its competitors (UNFCCC, 2023).

B2B firms' performance in emerging nations, such as India, is already being impacted by efforts to mitigate climate change. For instance, the Indian government has pledged to construct 450 GW of renewable energy capacity by 2030, presenting an enormous opportunity for B2B firms in the renewable energy sector, where more jobs will be created, and economic growth opportunities will arise (NRDC, 2022). In addition, B2B firms that already make eco-friendly efforts receive recognition. For example, Tata Steel Limited, a major steel producer in India, was recently named the "Global Slag Company of the Year" for its commitment to sustainable business practices and reducing greenhouse gas emissions (Steel, 2019). These examples show how B2B firms may gain from addressing climate change in fast-growing economies such as India. In addition, B2B firms' enhanced performance due to climate change efforts may result in improved sustainable competitiveness (Hostetter & Winkler, 2022). The term "sustainable competitiveness" refers to a company's capacity to fulfil the demands of current shareholders without jeopardising future generations' ability to do the same (SolAbility, 2022). Companies that care about the environment and work to mitigate climate change will be in a stronger position to adapt to the demands of today's consumers, investors, and government agencies (Coppola & Blohmke, 2019). These companies may improve their brand image, risk management, and access to new markets and income streams by lowering their environmental impact (OECD, 2020). In doing so, we cannot ignore the significant potential of digital technology, such as artificial intelligence (AI), or its associated process, such as artificial intelligence-based partner relationship management (AIPRM). AIPRM is a cutting-edge field that is transforming how businesses handle their relationships (Chatterjee, Chaudhuri, Vrontis, & Kadić-Maglajić, S., 2023). AIPRM can help businesses expedite their partner interactions, information processing, and overall performance by using AI algorithms and data analytics (Chatterjee et al., 2023). The ability of AIPRM to boost businesses' data processing is one of its primary advantages. Data input and analysis in conventional partner management systems are often performed manually, which can be a tedious and error-prone process (Chatterjee et al., 2023). However, AIPRM can automate many of these procedures, resulting in more streamlined data collection, analysis, and application for organisations. AIPRM, for instance, may aid businesses in tracking partners' performance trends, keeping tabs on the state of the industry, and developing strategies based on concrete data (Chatterjee et al., 2023). Furthermore, to improve firm performance and sustainable competitiveness, AIPRM may boost a firm's partner engagement and firm information processing capabilities for climate management in B2B firms. Concerning combating climate change, collaborative efforts between businesses have the potential to provide significantly more beneficial results than individual firms' unilateral monitoring of emissions motivated simply by self-interest (Dahlmann & Roehrich, 2019). As Paulraj, Chen, and Blome (2017) point out, firms' passionate pursuit of legitimacy and conformance with stakeholder norms may also be motivated by considerations of relationships. Furthermore, AIPRM may aid businesses in foreseeing and resolving

partnership challenges by offering timely feedback and insights (Chatterjee, Tamilmani, Rana, & Dwivedi, 2020). Through AIPRM, businesses may learn more about their partners' wants and requirements, leading to more fruitful cooperation and stronger partnerships (Chatterjee et al., 2023). AIPRM may aid businesses in responding to changing market circumstances and gaining competitive advantage by facilitating data-driven decision-making, optimising resource allocation, and fostering more effective collaboration with partners (Chatterjee et al., 2023). AIPRM can streamline partner management processes, improve data analytics and decision-making, and enhance collaboration and communication between firms and their partners (Dasanayaka, Al Serhan, Glamboosky, & Gleason, 2020). AIPRM can further improve firm performance and sustainable competitiveness by improving these areas. However, to fully realise the benefits of AIPRM, firms must ensure that their AIPRM systems are aligned with their overall strategy and goals, invest in the necessary technology and infrastructure, and develop a culture of innovation and continuous improvement. However, certain factors, such as information and communication technology, firm fit, and technological readiness, are necessary for the efficient execution of AIPRM. AIPRM calls for the acquisition, use, and analysis of enormous volumes of data, and to support it, the company must have a solid information and communication technology infrastructure (Chatterjee, Rana, Tamilmani, & Sharma, 2021), such as the equipment required to support data-intensive requirements of AIPRM. Furthermore, AIPRM must align with the company's strategic objectives, business model, culture, and resources, which is why firm fit is important (Chatterjee, Rana, et al., 2021). AIPRM may not connect with the firm's goals and may not provide the anticipated advantages if it is not a good fit. Similarly, the company must have technological readiness to implement and integrate AIPRM successfully (Chatterjee et al., 2023). This entails determining how prepared the company is to embrace and incorporate new technology. The company must determine whether it has the technological know-how, infrastructure, and resources needed to adopt and implement AIPRM in its operations.

Therefore, we employed the theoretical lenses of the dynamic capability view (DCV) and the resource-based view (RBV) to investigate the relationship between information communication technology, firm fit, technological readiness, AIPRM, firm information processing for climate management, firm partner engagement for climate management, firm performance, and sustainable competitiveness in the context of climate management measures implemented by B2B firms. The DCV highlights the significance of a company's adaptability to new circumstances and technologies, such as adopting AIPRM (Chatterjee et al., 2023). Companies need to use innovative methods to lessen their impact on the environment and advance sustainable growth due to climate change.³ By enhancing information processing and partner engagement for climate management, AIPRM may help businesses become more competitive over the long term. The RBV highlights firms' distinctive assets, which may be used to maintain a lead in the market. Using their technical resources to enhance information processing and partner engagement activities, businesses that use AIPRM may achieve a competitive advantage (Haefner, Wincen, Parida, & Gassmann, 2021). This has the potential to boost business performance and ensure long-term competitiveness.

In addition, the article is structured as follows: Section 2 explores the theoretical underpinnings of DCV and RBV along with developing hypotheses to investigate the relationship between different factors. Section 3 revolves around the methodology used in this research, followed by Section 4, where we have conducted the data analysis of this research. Furthermore, in Section 5, we discuss the findings of this research, followed by Section 6, where we offer different theoretical and practical implications. Finally, in Section 7, the conclusions, limitations,

² <https://www.unilever.com/planet-and-society/climate-action/de-carbonising-our-business/> (accessed on 20/04/2023)

³ <https://www.oecd.org/g20/summits/osaka/OECD-G20-Paper-Innovation-and-Green-Transition.pdf> (accessed on 21/04/2023)

and future research directions of this research are given.

2. Theoretical foundation and hypothesis development

2.1. Dynamic capability view (DCV) theory

This research incorporates the dynamic capability view (DCV) theory (Teece, Pisano, & Shuen, 1997) to determine what influence the uptake of an AIPRM solution throughout a sales channel has, as well as the effects of this uptake. According to Teece (2014), a dynamic capability is conceptualised along three dimensions: “sensing ability, seizing ability, and transforming/reconfiguring ability”. Therefore, businesses may use AIPRM as an external asset or an opportunity to meet ever-changing market demands by adopting a “Partner Relationships Management” solution (Chatterjee et al., 2023). However, this goal may not be met without external resources. Companies should also be able to assess the value of new possibilities and resources swiftly, reorganising and coordinating internal and external competencies to meet the challenges of changing business contexts. After absorbing new possibilities, businesses must make the most of them. For businesses to successfully integrate AIPRM throughout their sales channels, they must first cultivate many competencies (i.e., “sensing, seizing, and transforming or reconfiguring”) in their collaborators (Schreyogg, G. and Kliesch-Eberl, M., 2007). Accordingly, the partner relationship management operations of a company's partners' end-customers can be aided by AI, and the company, in turn, can offer its partners real-time, personalised services based on the information it gleans from the partner relationship management operations activities using AI (Chatterjee et al., 2023). As a result, collaborators will be able to better connect with their target audience. One way to do this is to partner with suppliers to better discover, evolve, and co-create worth for end customers. This is the capacity for perception (Chatterjee et al., 2023). Collaborators also require the capacity to gather and use the resources necessary to meet emerging demand in the marketplace quickly and effectively. Finally, businesses should learn to pool their resources in new ways to adapt to shifting markets (Fainshmidt, Pezeshkan, Lance Frazier, Nair, & Markowski, 2016).

2.2. Resource-based view (RBV) theory

The resource-based view (RBV) is a management concept for identifying the strategic assets needed by businesses to gain a market advantage (Barney, 1991). The RBV emphasises a company's assets to identify the skills and knowledge that may boost performance and lead to competitive advantage (Aker, Wamba, Gunasekaran, Dubey, & Childe, 2016). Combining many organisations' potentially static and diverse resources is foundational to the concept of competitive advantage in B2B relationship administration (Barney, 1991). The primary focus of RBV theory is on how well businesses can manage their resources while working together in a B2B setting (Kozlenkova, Samaha, & Palmatier, 2014). Different organisations function differently because they have different strengths and ways of handling resources (Short, 2014). The RBV suggests that businesses should share information and ideas to strengthen their B2B ties (Short, 2014). Employees from diverse companies with B2B ties benefit from an open exchange of ideas, knowledge, and experience (Crick & Crick, 2020). Improving B2B connections is possible with appropriate AIPRM deployment and utilisation (Libai et al., 2020). As a bonus, this will make it easier to realise the full capability of the technology deployed inside the business (Mikalef, Boura, Lekakos, & Krogstie, 2019). Corporate performance and advancement in a B2B scenario, as measured by RBV, is contingent on the company's inventory of internal resources, such as material capital, personnel capital, and organisation capital (Barney, 1991).

After conducting an extensive literature review, we present in Table 1 the relevant prior studies, where we can see how previous scholars have addressed the climate action measures used in various

Table 1

Recent studies on climate actions in B2B firms.

S. No.	Author(s)	Research objectives	Contributions to the literature
1	Kim, Atukeren, and Kim (2023)	This research looks at how greenhouse gas emissions affect the monetary performance of B2B firms.	Their research shows that cutting down on greenhouse gas emissions is an accepted expense for B2B companies. Furthermore, the findings suggest that stricter regulation may be needed at the policy level for B2B enterprises to cut their greenhouse gas emissions.
2	Rahman, Bag, Gupta, and Sivarajah (2023)	The authors look into how technological readiness at B2B enterprises affects their information technology prowess, AI-based customer relationship management, and their capacity for relationship building and social responsibility.	The research, grounded on the concept of dynamic capability view, revealed a positive relationship between a B2B company's technological readiness and its information and communication technology and AI-based customer relationship management capability. The capacity of a business to use information and communication technology as a mediator between technological readiness and AI-based customer relationship management capability is a major conclusion of this research.
3	Chatterjee et al. (2023)	Investigates the factors influencing the adoption of AIPRM by B2B companies and the resulting value to the company.	This research leverages the dynamic capability view and absorptive capacity theory to provide a conceptual model for determining what it takes for a B2B company to embrace AIPRM and how that adoption will affect the firm's bottom line. The findings demonstrate that the implementation of an AIPRM system resulted in enhanced operational performance, corporate value creation, and enhanced partner services and engagement.
4	Vallaster, Cesinger, Niemand, Huis, and Lienbacher (2023)	The authors utilise social identity theory and seminal work to understand the inclinations of decision-makers in forming B2B relationships.	This study provides a nuanced outlook on the micro-foundations of B2B relationships and how individual social identity styles influence the inclination to participate in B2B interactions.
5	Chatterjee, Chaudhuri, and Vrontis (2022)	The research aims to learn how B2B relationship management may benefit from using an AI-based customer relationship management system.	The research shows that using AI-based customer relationship management significantly improves the quality of B2B relationships and the overall success of businesses.
6	Dwivedi and Wang (2022)	The research aims to investigate the motivation of B2B firms	This study investigates the motivations behind B2B organisations' use of AI for marketing

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Table 1 (continued)

S. No.	Author(s)	Research objectives	Contributions to the literature
		to use AI in their marketing strategies.	endeavours. It explores the potential applications of AI in fostering innovation and optimising supply chain networks. Additionally, it examines how AI may boost B2B customer experience and customer relationship management. Last, it delves into the utilisation of AI in developing dynamic capabilities within B2B marketing strategies.
7	Chatterjee, Rana, et al. (2021)	This research explores the potential effects of AI-based customer relationship management on company success across a spectrum of company size, age of firm, and industry.	This research used two well-established theories as theoretical lenses, such as institutional theory and the resource-based view, to evaluate the effect of AI-based customer relationship management on the performance and competitive advantage of B2B organisations. According to the findings, the quality of the AI-based customer relationship management implementation process is the most important factor in determining the success or failure of AI-based customer relationship management deployment in B2B organisations.
8	Saura, Ribeiro-Soriano, and Palacios-Marques, D. (2021)	The study investigates the role of customer relationship management in digital B2B marketing.	This research compiles a review of 30 scholarly articles on the topic of customer relationship management systems and AI-based customer relationship management in B2B settings. These papers have been carefully analysed to delve into the study's goals. The findings of the study categorise a variety of customer relationship management systems and their classifications, as well as investigate the primary methodologies and applications of AI-based customer relationship management in B2B digital marketing.
9	Present Study	Investigate the relationship between information and communication technology capability, firm technological readiness, firm fit, AIPRM, firm partner engagement, firm information processing, firm performance, and sustainable competitiveness to	Drawing on the dynamic capability view and resource based view, the present research indicates that, information and communication technology capability, firm technological readiness, and firm fit are prerequisite to implementing AIPRM as all three of them positively affects AIPRM.

Table 1 (continued)

S. No.	Author(s)	Research objectives	Contributions to the literature
		understand the role of these factors in climate-based actions within B2B firms.	Furthermore, implementation of AIPRM, improves the firm partner engagement and firm information processing, leading to better firm performance and sustainable competitiveness.

firms, as well as the major contributions that have been identified to offer the originality of this study through its novel contribution to the literature.

2.3. Hypothesis formulation and model development

To get an edge in the market, companies must combine their information and communication technology capabilities with other resources, both material and immaterial (Aker, Gunasekaran, Wamba, Babu, & Hani, 2020). Building new capabilities requires several steps, including information collection, information security, information dissemination, and data-driven decision-making. In addition, companies should reorganise and adapt their resources to maximise potential gains during times of uncertainty (Sunday & Vera, 2018). This is crucial for establishing information and communication technology and paving the way for the growth of AIPRM capability (Chatterjee, Rana, et al., 2021). Furthermore, the incorporation of big data analytics and AI functionality into B2B firms has been made possible by advancements in information and communication technology capabilities (Chung, Kingshott, MacDonald, & Putranta, 2021). In addition, the establishment of AIPRM capabilities necessitates the creation of a proficient and flexible information technology planning procedure through cooperative efforts with industry associates. It is crucial to have sufficient information technology infrastructure to provide seamless operations and services (Rahman et al., 2023). Since AIPRM is conceived of as a dynamic capacity, it has its roots in DCV. This is because it is based on the alteration of routines, such as the development of information and communication technology. To thrive in today's fast-paced business environment, companies must be able to rapidly and efficiently realign their distinctive assets and competencies to seize opportunities and satisfy customer needs. Additionally, B2B firms must provide an environment that encourages their industrial collaborators to experiment with novel uses of information and communication technology (Lu & Ramamurthy, 2011). Hence, we hypothesise:

H1. *The information and communication technology capabilities of B2B firms have a positive relationship with AIPRM capabilities.*

For companies to successfully adopt new Industry 4.0 technologies, their technological readiness is crucial (Telukdarie, Buhulaiga, Bag, Gupta, & Luo, 2018). According to Chatterjee (2019), AI is essential for supporting digital partner relationship management activities inside the company. AIPRM is conceptualised as a dynamic capability because it is based on shifting routines, such as demonstrating innovation in technological readiness and working towards increasing information and communication technology competence, both of which have their origins in DCV. "Dynamics 365 for customer insights" is one example of a system with such intelligence. Data sets from customers may be analysed using this technology for prediction purposes. Chatterjee (2019) identify many firms, including "Zoho, Sugar PRM, and Sales Force", that provide AIPRM technologies. To further expand information and communication technology capabilities, technological readiness necessitates the contribution of several essential resources (Gupta & George, 2016). The intended outcomes in terms of productivity enhancement can be

achieved via the implementation of regular digital technology training and education. B2B firms need to make financial investments in cutting-edge information and communication technology training, as well as the construction of technical infrastructure, to implement new technologies effectively (Chatterjee, Rana, et al., 2021). In addition, Chatterjee (2019) note that organisations can experience difficulties with infrastructure when implementing AIPRM technologies, difficulties that may be solved with a ready plan and good training. In addition, Chatterjee, Ghosh, Chaudhuri, and Chaudhuri (2021) argue that nine factors are necessary for the acceptance of an AI-integrated partner relationship management-knowledge management structure: “design and development of the AI-integrated partner relationship management-knowledge management tool”, “support of the immediate manager”, “adequate training and readiness”, “business value addition”, “adequate security mechanism”, “developing a privacy policy for the AI-integrated partner relationship management-knowledge management”, “simplicity of the new AI-integrated partner relationship management-knowledge management system”, “supporting legal requirements” and “ease of use”. The aforementioned research provides sufficient evidence that training and preparation significantly impact AIPRM. In this vein, we propose that companies' technological readiness is a key factor in developing their capacity for AIPRM (Chatterjee et al., 2023). Therefore, we hypothesise the following:

H2. *The level of technological readiness in B2B firms has a positive relationship with their level of AIPRM capabilities.*

Research into the accounting and information systems model reveals that the level of technological proficiency already present in an organisation (also called “firm fit”) is the result of a complex interaction between system features and organisational characteristics (Chatterjee, Rana, et al., 2021). In the scenario of introducing new technology, the four main characteristics of organisations may also be used to conceptualise firm fit. These constituent parts are structure, function, technology, and people (Chatterjee et al., 2023). As DCV implies, a dynamic business fit or environment must be present. When creating AIPRM, businesses must acquire and use resources to adapt to changes in the business environment over time (Makadok, 2001). In addition, AIPRM may be directed by firm fit if it is to aid businesses in implementing competitive tactics because market circumstances are always shifting via the creative combination and transformation of existing resources (Eriksson, 2014). After looking at these arguments, one may conclude that organisational readiness aids in successfully introducing a new system. This leads to the following hypothesis:

H3. *Firm fit has a positive relationship with implementing AIPRM in B2B firms.*

Collaboration among enterprises has the potential to provide much more favourable outcomes in addressing the issue of climate change compared to the unilateral monitoring of emissions only driven by a firm's self-interest (Dahmann & Roehrich, 2019). In addition, businesses may have relational motivations for aggressively pursuing legitimacy and conformity with stakeholder standards (Paulraj et al., 2017). Business strategies, activities, and business settings in which supply chain involvement is carried out are all intertwined with a company's motivations for implementing carbon reduction initiatives, whether ethical, interpersonal, or functional (Lintukangas, Arminen, Kähkönen, & Karttunen, 2022). Data availability in Industry 4.0 fuels AI-driven applications such as partner relationship management and enables affordable, segmented service delivery at scale (Chatterjee, Rana, et al., 2021). According to Bag, Gupta, Kumar, and Sivarajah (2021), AI systems driven by large data sets may help businesses better understand their clients, employees, and competitors. As a result, AIPRM systems may aid in vetting high-value clients, forecasting future business, and allocating resources accordingly. Since B2B partners (i.e., supplier and buyer companies) may wish to remain affiliated and in frequent contact to seek out prospective business possibilities, this will help to boost

relationship performance. According to Chatterjee, Rana, et al. (2021), partner relationship management quality and satisfaction greatly affect workers' opinions and propensities to use AIPRM solutions. The RBV holds that a company's competitive edge and long-term viability improve when it has access to scarce, high-value resources that are difficult for rivals to replicate (Short, 2014). Everyone benefits when companies collaborate to improve supply chain capabilities by investing time, money, and expertise (GEP, 2021). Interfirm competitive advantage is based on specialised resources, knowledge-sharing procedures, complementary endowments and competencies, and sound governance practices that firms develop and cultivate via their interactions with one another (Rahman et al., 2023). According to Chung, Kingshott, MacDonald and Putranta (2021), the capacity for mass customisation is positively associated with the analytical intelligence provided by big data, improving partner relationship management performance. This will eventually improve partners' engagement with end customers. Improving collaborators' capacities for value identification, development, and cocreation with suppliers may lead to better, more individualised services for end consumers (Chatterjee et al., 2021, b). This is the capacity for perception. However, there is a lack of literature on how AIPRM capabilities relate to B2B firms' partner engagement, especially for climate management. Thus, we propose the following hypothesis:

H4. *AIPRM capabilities positively affect B2B firms' partner engagement in climate management.*

Since the transition from competitive to cooperative B2B partnerships is hampered by information asymmetry, businesses require facilitation skills to sustain collaborative B2B connections (Cherbib, Chebbi, Yahiaoui, Thrassou, & Sakka, 2021). These enabling skills, also known as competencies of the company, include information technology-based applications and technologies. In a B2B setting, workers have been observed to use various methods for strengthening ties with their counterparts at other companies. An AIPRM system may help improve these methods (Chatterjee et al., 2020). If these methods are determined to be ineffective, companies may find it challenging to continue working together effectively. As a result, B2B interactions are impeded (Demlechner, Schoemer, & Laumer, 2021). Climate management is possible through firms' information processing by monitoring environmental standards, and it is believed that B2B information processing systems may be enhanced by installing an AIPRM platform for administering B2B relationships (Lacka, Chan, & Wang, 2020). Furthermore, AI-facilitated collaborative administration should incorporate the capacity to merge assets to innovate and adapt to shifting market circumstances, therefore enhancing the efficacy of information sharing (Fainshmidt et al., 2016). From the DCV viewpoint, AIPRM implementation across the sales network develops various capabilities (i.e., sensing, seizing, and transforming or reconfiguring) of partners for adopting AIPRM systems based on effective information sharing (Rahman et al., 2023). This is the capability of converting or resetting based on DCV. Accordingly, we can hypothesise as follows:

H5. *AIPRM capabilities positively affect B2B firms' information processing systems for climate management.*

Researchers examining firm partnerships argue that, as a result of resource and capability constraints, the acquisition of external resources significantly influences firm performance (Chuang, Dahlin, Thomson, Lai, & Yang, 2015). Furthermore, they increasingly assert that exceptional performance is not solely dependent on the success of individual partnerships but rather on the collective impact of the entire portfolio of collaborations (Wassmer, 2008). Furthermore, environmental communication strategies aim to demonstrate to the public and the media that environmental problems are given significant attention by management. By doing so, it fosters strong and cooperative connections between the business and its stakeholders (Nair & Menon, 2008). Therefore, a strong environmentally friendly corporate image serves as a distinct indication of a company's dedication to the environment for its primary

stakeholders and serves as an efficient method to enhance the company's reputation, environmental credibility, environmental differentiation strategy (Heikkurinen, 2010), and financial success. In this context, one of the organisational factors that contribute to the profitability of environmentally friendly corporations is the establishment of a reputation for leadership in environmental matters or a positive image in terms of environmental consciousness (Martín-de Castro, Amores-Salvadó, & Navas-López, 2015). This reputation can effectively enhance sales among environmentally conscious customers, providing the firm with a competitive edge over its rivals in the market (Martín-de Castro et al., 2015). B2B partnership channels provide access to outside assets that are complementary to an organisation's needs, and the alliances formed between vendors and suppliers to provide these resources directly affect an enterprise's success (Cainelli, De Marchi, & Grandinetti, 2015). Utilising cutting-edge technology to engage with their suppliers and consumers, businesses can take advantage of dynamic capabilities by continually developing, modifying, and redesigning internal and external competencies, supporting partnership management and reaching firm performance (Rahman et al., 2023). Thus, it can be concluded that overall business success is significantly influenced by factors such as the efficiency of B2B interaction strategies, the expertise of the workforce, and the efficacy of information processing systems. Based on these considerations, the following hypothesis is proposed:

H6. *B2B firms' partner engagement in climate management positively affects firm performance.*

In this context, “process” refers to the unique set of inputs, activities, participants, and outputs that may be measured. In the B2B sphere, data processing systems are a vital resource (Dwivedi, Ismagilova, Rana, & Raman, 2021). In addition to facilitating the identification of particular areas that obstruct information flow among the organisations engaged in B2B relationship management, an information processing system gives a calibrated measure of variability for the output of a given process (Farrokhi, Shirazi, Hajli, & Tajvdi, 2020). Information processing systems are capable of generating valuable data about climate management. These data encompass various aspects, such as partner environmental performance, emissions data, energy usage, and compliance with sustainability standards. DCV aligns with this model, indicating that the corporate setting is dynamic. To adapt to changing market conditions, businesses must invest and allocate assets (Makadok, 2001). While information technology capabilities may help businesses execute competitive objectives while considering shifting market circumstances, these capabilities are also dynamic, as they can combine and transform the firm's available resources in a novel way to achieve firm performance (Eriksson, 2014). Organisations with well-functioning information technology are expected to gain positive B2B relationship management results after installing an AIPRM solution (Dwivedi et al., 2021). Consistent with RBV's central tenet, which states that improved performance results from enterprises pooling resources and information to create a more cohesive supply chain, this strategy emphasises the need for both parties to invest in the development of its members (Rahman et al., 2023). The above discussion leads to the formulation of the following hypothesis:

H7. *B2B firms' information processing systems for climate management positively affect firm performance.*

Organisational performance is evaluated in terms of the net results the organisation produces. Financial and promotional factors, including business share, return on investment, growth in sales, and net revenue, are all taken into account (Kreye & Perunovic, 2020). In addition, the degree of satisfaction is used as a reliable indicator of the efficiency of an organisation (Keramati, Mehrabi, & Mojir, 2010). The subjective assessment of an organisation's ability to successfully manage its B2B relationships is correlated with the performance of organisations functioning in a B2B setting (Sin, Tse, & Yim, 2005). Organisational success is largely determined by the means through which it accumulates

“valuable, inimitable, rare, and nonsubstitutable” resources; as stated, RBV theory is a useful tool for determining these means (Chatterjee, Rana, et al., 2021). It is believed that these capabilities affect an organisation's ability to maintain its sustainable competitive advantage. By accomplishing these goals, organisations are seen as gaining a competitive edge over other existing contemporaneous organisations that are already operating in the market (Rahman et al., 2023). The extent to which the performance of an organisation operating in a B2B setting is capable of achieving higher advantages in terms of social well-being in comparison to the performance of other organisations operating under circumstances that are comparable to those conditions is the definition of sustainable competitive advantage (Chatterjee, Rana, et al., 2021). The above discussion leads to the formulation of the following hypothesis:

H8. *Firm performance positively affects firms' sustainable competitiveness.*

Based on the above hypotheses, a hypothesised model is proposed, as shown in Fig. 1.

3. Research methodology

3.1. Context, hypothesis validation, and questionnaire design

The recommended three-stage study design was employed in this research (Schilke, 2014). Fig. 2 depicts the research process followed for this study.

The research consisted of in-depth interviews with those who are working in B2B firms. During the first section of the interview, questions were asked regarding various topics related to their corporate climate action capabilities, from the perspective of firm fit, information and communication technology capabilities, and firms' technological readiness for AIPRM. We also asked them to explain the role of AIPRM in firm partner engagement and firm information processing for climate management and its cascading effect on firms' performance and longevity based on taking actions oriented towards climate change. The second component of the study follows naturally, critically examining the study's hypotheses through interviews with business leaders about the importance of various firm capabilities for adopting and effectively implementing AIPRM, which ultimately results in sustainable competitive advantage. There was widespread agreement that organisations' capabilities can facilitate effective AIPRM implementation and progress towards firm performance and sustainable competitiveness. We have confidence in our hypotheses since the academic literature supports them before moving on to the third and last section of the pilot phase interview. In this section, the comments on the clarity of the questions and their challenges in responding were collected, having filled out a draft version of the questionnaire to be used in future survey research. Several questions were either reworded or removed as a direct consequence of this procedure. Ultimately, we acquired data related to the constructs of our research from the relevant organisations and performed analysis on that data.

3.2. Research survey

Initially, the authors posited a hypothesised model derived from the literature and brainstorming within the research team members. Subsequently, this model was used as a guiding tool for the study settings. Ultimately, the research uses these parameters to direct the structures that the authors choose to investigate. To ensure the accuracy of the authors' questionnaire, a preliminary testing phase was conducted with area experts who have experience in organisations that have implemented climate-based activities, such as the adoption of green technology and the integration of AI-facilitated services into their B2B operations. The researchers sought the valuable input of industry specialists to ensure that the phrasing chosen for each of the thirty-four items would not be seen as unclear by the respondents. Several

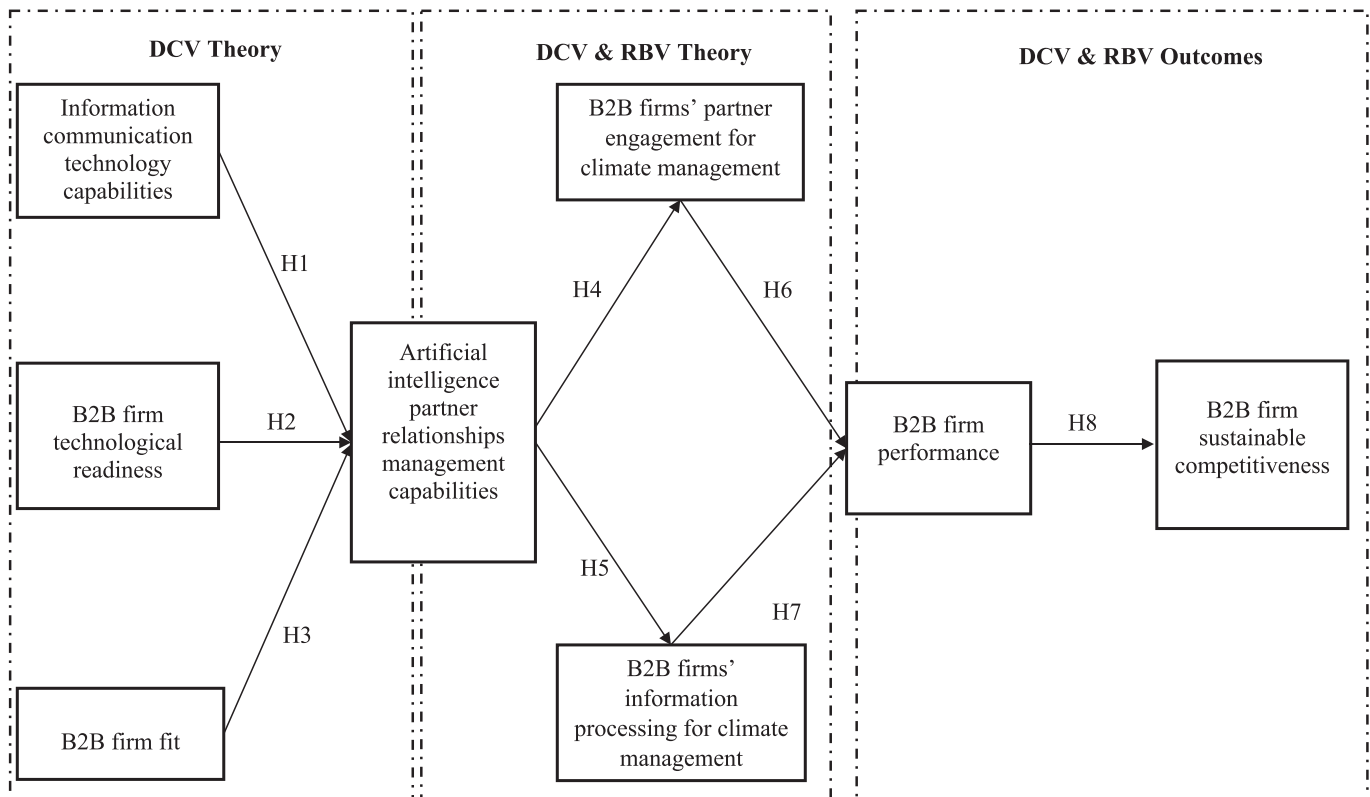


Fig. 1. Proposed hypothesised model.

suggestions and feedback were found to be truly valuable, especially concerning the exercises that the authors needed to repeat. Based on the recommendations, the selected items were used as follows: information and communication technology capabilities (4 items), B2B firm technological readiness (4 items), B2B firm fit (4 items), artificial intelligence partner relationship management (4 items), B2B firms' partner engagement for climate management (4 items), B2B firms' information processing for climate management (4 items), B2B firm performance (5 items), and B2B firm sustainable competitiveness (5 items). Additionally, the authors assigned codes to these items to enhance their presentation, as shown in Table 2. The researchers used a structured study questionnaire to gather data from organisations that have implemented climate-focused initiatives, including the adoption of environmentally friendly technology and the integration of AI-enabled services into their B2B firms' operations. By gathering information from a wide range of B2B firms from an emerging economy, i.e., India, we hoped to lessen the possibility of skewed results and draw broader conclusions from our analysis. The data were collected from January to March of 2023, focusing on Indian B2B firms that use information and communication technology capabilities for decision-making related to AIPRM. Acquiring data from enterprises in India might present difficulties, especially given the lack of established professional networks within the sector. Therefore, to elicit responses from the participants, the authors used several sources, including social media platforms, official government websites, and personal networks.

The research area of this study is current and new, therefore with the chain referral process, we reached out to firms who possess a keen interest and expertise in understanding the significance of information and communication technology capabilities, as well as AIPRM, and their impact on organisational performance and the attainment of a sustainable competitive advantage. The chain referral strategy is used to capture the network structure while preserving the relevant information (Kolaczyk, 2009). The process involves iteratively gathering information from vertices that are connected to the vertices acquired in the

preceding iteration (Chan, 2020). Additionally, it is a cost-effective approach for sample collection. Academic researchers can efficiently acquire substantial quantities of data by gathering information from the vertices that are connected to the samples in the preceding iteration (Chan, 2020).

After reaching the firms, we connected with the people and explained the objectives of the study to them so that easy for them to understand and participate. By implementing random sampling, an equal chance to everyone was given to participate in the survey. This approach aims to create a representative sample that accurately reflects the characteristics of the complete population. In this way, we were able to connect with the respondents for unbiased responses. Furthermore, the participants played a vital role in driving their respective organisations' digital transformation initiatives at various hierarchical levels. The participants were provided with the requisite technological tools to implement AIPRM and use it to enhance organisational performance and gain a competitive edge in the marketplace. Given these considerations, we are certain that these managers will offer an accurate and relevant assessment of the aims of the current study. We finally received a total of 142 responses from B2B firms for further data analysis.

3.3. Nonresponse bias

We looked for signs of nonresponse bias using a *t*-test. We used the *t*-test on two sets of data, the first from January 2023 to the middle of February 2023 and the second from the middle of February 2023 to the end of March 2023. According to this two-set analysis, there are no statistically significant variations ($p > 0.05$) between initial and later respondents regarding the means of any of the theoretical constructs tested (Armstrong & Overton, 1977). Table 3 details the types of businesses that responded to our survey.

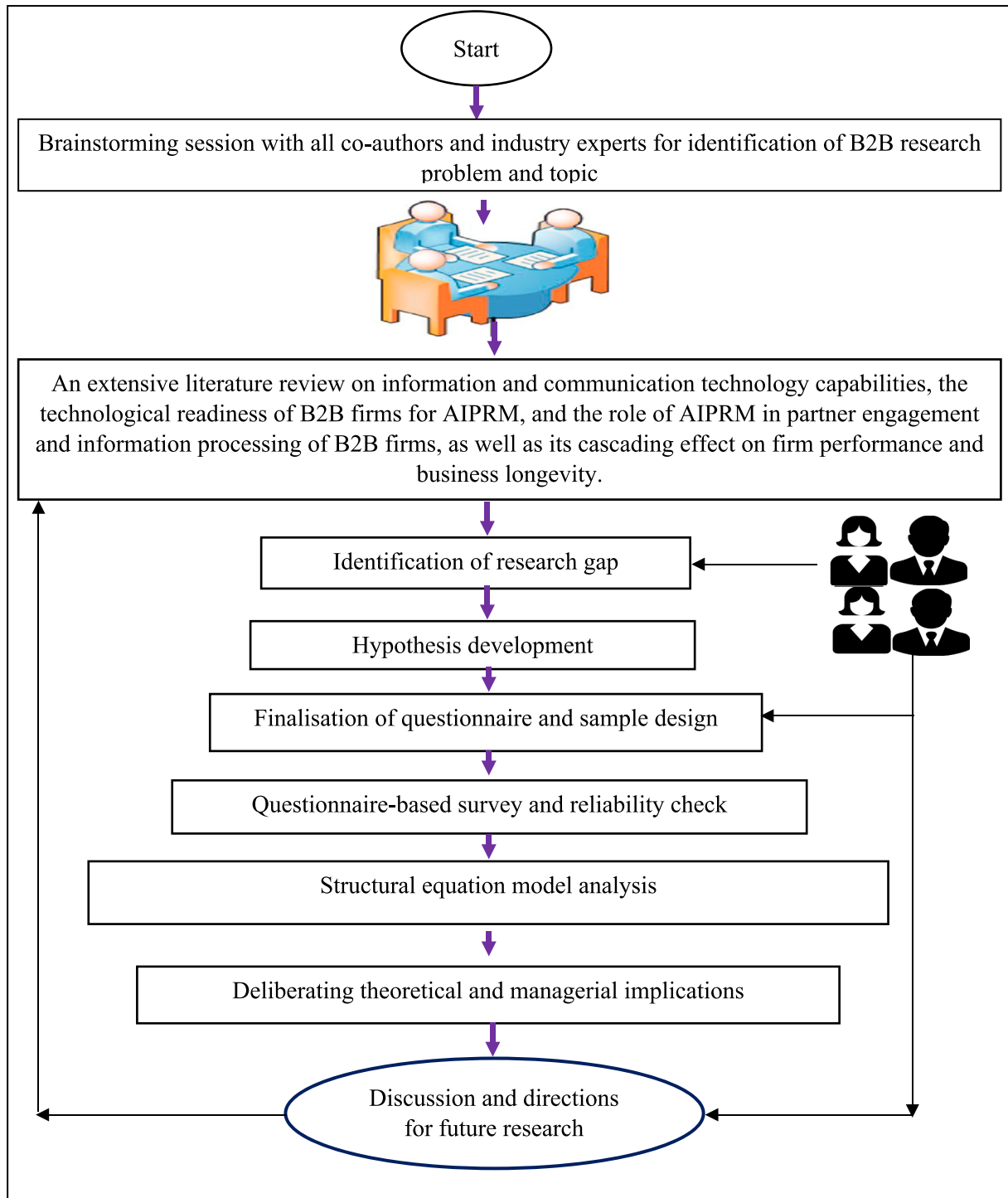


Fig. 2. Research process flowchart.

4. Data analysis

4.1. Common method bias (CMB)

The introduction text used in a survey may steer responses from various respondents in the same way, leading to some overlap in the indicators' variance. The implied social acceptability of a certain survey response might also contribute to CMB by encouraging respondents to respond to questions consistently. The complete collinearity test was introduced by [Kock and Lynn \(2012\)](#) as an all-encompassing method for evaluating collinearity in both the vertical and horizontal planes

simultaneously. This method, which is carried out in its entirety by the SmartPLS software, produces variation inflation factor values for all latent factors in a model. If the variation inflation factor is more than 3.3, it is suggested that CMB may affect the model and may indicate severe collinearity ([Kock, 2015](#)). Therefore, we can say that the model is CMB-free if every variation inflation factor value calculated as part of a comprehensive collinearity test is less than or equal to 3.3 ([Kock, 2015](#)). We calculated the variation inflation factor value using SmartPLS software and obtained values ranging from 1.000 to 2.550 for all the latent variables, which is less than the threshold of 3.3. Based on these findings, we can state that this study has no issues with CMB.

Table 2

Constructs, items, and brief descriptions.

Constructs	Item	Description	Sources
Information and communication technology capabilities (ICTC)	ICTC1	Our streamlined services and operations are made possible by our cutting-edge information and communication technology infrastructure.	Stoel and Muhanna (2009); Lu and Ramamurthy (2011); Rahman et al. (2023)
	ICTC2	Together with our business collaborators, we have developed a method for strategic information and technology planning that is both effective and adaptable.	
	ICTC3	Our company is committed to keeping its industrial collaborators abreast of the latest developments in information technology.	
	ICTC4	Our company fosters an environment that encourages our business associates to investigate innovative ways to use information and communication technology.	
B2B Firm technology readiness (FTR)	FTR1	At our company, we deliver frequent training on cutting-edge digital technology and highlight its advantages to our staff.	Vize, Coughlan, Kennedy, & Ellis-Chadwick, 2013; Rahman et al. (2023); Chatterjee et al. (2023)
	FTR2	Our company has spent a lot on developing the necessary information and technology systems for adopting data-driven AIPRM.	
	FTR3	Our company has been ahead of the curve in adopting cutting-edge digital technology.	
	FTR4	The employment of digital technology in the workplace is more effective than traditional methods.	
B2B Firm fit (FF)	FF1	When using AIPRM, organisations should conduct pilot projects to assess whether the technology is a good fit for them.	Chatterjee, Rana, et al. (2021); Chatterjee et al. (2023)
	FF2	For effective B2B relationship management, firms should adopt an AIPRM approach.	
	FF3	To successfully use AIPRM for B2B relationship	

Table 2 (continued)

Constructs	Item	Description	Sources
Artificial intelligence partner relationship management capabilities (AIPRM)	FF4	administration, an organisation needs access to the right data. AIPRM must be easily integrated into the organisation's existing IT infrastructure.	Lacka et al. (2020); Chatterjee et al. (2023)
	AIPRM1	Integrating AI into the current partner relationship management platform improves collaborators' relationship management processes.	
	AIPRM2	Customers that utilise AIPRM will have access to sufficient support personnel.	
	AIPRM3	To effectively integrate AIPRM for B2B relationship administration, our company has devoted enough to expenditure and resources.	
B2B firms' partner engagement for climate management (FPE)	AIPRM4	Since using AIPRM, employee satisfaction has increased across our company.	Sirdeshmukh, Singh, and Sabol (2002); Lages, Lancastre, and Lages (2008); Rahman et al. (2023)
	FPE1	We keep in close contact with our current business associates.	
	FPE2	Our current industrial collaborators are the ones we would select to work with again if we were to start this company over.	
	FPE3	Our company's policies demonstrate our appreciation for our industrial clientele.	
B2B Firms' information processing for climate management (FIP)	FPE4	Our business collaborators are trusted confidants with whom we disclose proprietary information in exchange for similar disclosures.	Johnston and Cortez (2018); Lacka et al. (2020); Chatterjee, Rana, et al. (2021)
	FIP1	Implementing AIPRM has sped up and simplified B2B contact administration.	
	FIP2	Using AIPRM, we can send and receive orders with our collaborators considerably more quickly.	
	FIP3	We benefit from AIPRM because it provides instantaneous suggestions which were previously unavailable through conventional PRM tools.	
	FIP4	Our company is privy to the necessary data	

(continued on next page)

Table 2 (continued)

Constructs	Item	Description	Sources
B2B firm performance (FP)	FP1	sets to put AIPRM into practice.	Kreye and Perunovic (2020); Chatterjee, Rana, et al. (2021)
		We anticipate that the organisation's functional efficiency will increase because of the adoption of AIPRM, resulting in lower material costs.	
	FP2	We anticipate increased revenue growth because of better customer service after the introduction of AIPRM.	
		Our company's reputation has improved since we began using AIPRM for B2B relationship administration, which has led to a greater return on investment.	
	FP3	We feel that the speed with which AIPRM can make decisions benefits both the company's bottom line and the environment.	
		We believe that AIPRM is helping the company expand financially.	
	FP4	Our whole business strategy includes the use of AIPRM.	
		We believe that an advantage may be gained over the competition with the proper application of AIPRM.	
	FP5	We believe that AIPRM helps businesses enhance their climate and environment actions.	
		Our company is committed to implementing socially responsible business practices and associated initiatives.	
B2B firm Sustainable competitiveness (SC)	SC1	We believe that AIPRM may help reduce pollution levels in the atmosphere.	Sin et al. (2005); Kapitan, Kennedy, and Berth (2019); Chatterjee, Rana, et al. (2021)
	SC2		
	SC3		
	SC4		
	SC5		

4.2. Psychometric analysis

Following the guidelines set out by Anderson and Gerbing (1988), we analysed the measurement model's output to determine the psychometric features of our multi-item constructs. Therefore, content validity was achieved through literature-informed, methodical questionnaire preparation and the question items used to measure the constructs. Table 4 displays the estimated standard coefficients of the indicators used, showing that they each loaded on the intended construct, without any substantial cross-loadings. Discriminant validity was evaluated by looking at the correlation coefficients against the square root of the average variance extracted (AVE) scores for each construct. Table 5 shows that when using the Fornell and Larcker criterion (Fornell &

Table 3

Sample characteristics.

	Set 1 (January to mid-February) responses	Set 2 (Mid-February to March) responses	Total response
Industry			
IT services and consulting	9	14	16.20%
Logistics	13	18	21.83%
Finance and banking	19	21	28.17%
Manufacturing firms	17	31	33.80%
Gender			
Male	21	59	56.34%
Female	37	25	43.66%
Education			
Graduate	29	47	53.52%
Post-graduate	19	28	33.10%
PhD	10	9	13.38%
Geography			
East	4	13	11.97%
West	17	24	28.87%
North	11	16	19.01%
South	26	31	40.15%
Experience			
Up to 5 years	17	19	25.35%
5–10 years	24	39	44.37%
10–15 years	12	17	20.42%
More than 15 years	5	9	9.86%
Firm age (In terms of using AI)			
Less than 1 year	7	13	14.08%
1–5 years	33	41	52.12%
More than 5 years	18	30	33.80%

Table 4

Construct reliability, validity, model fit, and robustness.

SRMR = 0.07						
Constructs	Items	Loadings	α	ρ_{a_a}	ρ_{a_c}	AVE
AIPRM	AIPRM1	0.895	0.916	0.917	0.941	0.799
	AIPRM2	0.876				
	AIPRM3	0.898				
	AIPRM4	0.907				
FIP	FIP1	0.904	0.920	0.920	0.943	0.806
	FIP2	0.901				
	FIP3	0.898				
	FIP4	0.889				
FPE	FPE1	0.913	0.930	0.930	0.950	0.826
	FPE2	0.903				
	FPE3	0.896				
	FPE4	0.923				
FTR	FTR1	0.890	0.870	0.873	0.920	0.793
	FTR2	0.903				
	FTR3	0.879				
	FF1	0.892	0.900	0.903	0.930	0.769
FF	FF2	0.857				
	FF3	0.870				
	FF4	0.887				
FP	FP1	0.902	0.944	0.944	0.957	0.818
	FP2	0.898				
	FP3	0.903				
	FP4	0.881				
SC	SC1	0.903	0.938	0.939	0.953	0.802
	SC2	0.906				
	SC3	0.892				
	SC4	0.900				
ICTC	ICTC1	0.919	0.925	0.927	0.947	0.817
	ICTC2	0.888				
	ICTC3	0.900				
	ICTC4	0.907				

Table 5
Discriminant validity (Fornell-Larcker/HTMT).

Discriminant Validity (Fornell-Larcker criterion)								
	AIPRM	FIP	FPE	FTR	FF	FP	SC	ICTC
AIPRM	0.894							
FIP	0.651	0.898						
FPE	0.782	0.685	0.909					
FTR	0.759	0.591	0.676	0.891				
FF	0.731	0.613	0.733	0.736	0.877			
FP	0.637	0.743	0.685	0.593	0.587	0.904		
SC	0.663	0.776	0.717	0.609	0.597	0.776	0.896	
ICTC	0.746	0.607	0.657	0.677	0.656	0.584	0.589	0.904

Discriminant Validity (Heterotrait-monotrait ratio (HTMT) criterion)								
	AIPRM	FIP	FPE	FTR	FF	FP	SC	ICTC
AIPRM								
FIP	0.709							
FPE	0.846	0.740						
FTR	0.849	0.660	0.750					
FF	0.802	0.673	0.800	0.831				
FP	0.684	0.797	0.731	0.655	0.637			
SC	0.715	0.835	0.766	0.674	0.651	0.824		
ICTC	0.808	0.657	0.706	0.754	0.718	0.623	0.631	

Larcker, 1981), the former was always larger than the latter.

A lack of discriminant validity is not consistently detected by these methods (Fornell and Larcker criterion), as shown in a simulation study by Henseler, Ringle, and Sarstedt (2015). To evaluate discriminant validity, Henseler et al. (2015) offer a new technique based on the multitrait-multimethod matrix called the “heterotrait-monotrait (HTMT) ratio” of correlations. In addition, they use a Monte Carlo simulation analysis to show that the novel method outperforms the Fornell-Larcker criterion and the evaluation of (partial) cross-loadings. Therefore, we additionally evaluated discriminant validity using HTMT standards (as shown in Table 5). The consistency of our model is confirmed by both the Cronbach's alpha and composite reliability scores obtained. Construct validity was established after verification of adequate levels of “content, convergent, discriminant” validity and reliability (O’Leary-Kelly, Vokurka, & R., 1998). These analyses informed the conclusion that the constructs were adequately measured, allowing us to move on to hypothesis testing.

4.3. Structural model assessment

We evaluated the structural model by assessing the squared multiple correlations (R^2) of the endogenous variables, Cohen's f square of the exogenous variables, the impact of path coefficients, the predictive relevance (Stone-Geisser's Q^2 value) of the exogenous variable, and the evaluation of unobserved variation (Schoenherr, 2016). All of these factors were taken into consideration. Table 4 shows the R^2 values of each variable explained by the model. The model explains 70.2% of the variance in AIPRM, 42.4% of the variance in B2B firms' information processing for climate management (FIP), 61.1% of the variance in B2B firms' partner engagement for climate management (FPE), 61% of the variance in B2B firm performance (FP), and 60.3% of the variance in B2B firm sustainable competitiveness (SC).

In addition, we calculated the impact of the exogenous variable using Cohen's f^2 . This measure is used to assess the relative impact of a predictor construct on an endogenous construct in terms of its explanatory power. This can be done by comparing the two constructs. For this research, the range of the effect size is from 0.089 to 1.570, covering a wide spectrum of small to large effect sizes of various exogenous variables (Hair, Hult, Ringle, & Sarstedt, 2016). The Q^2 value proposed by Stone-Geisser was also evaluated. Predictive relevance, or Q^2 , is used to evaluate a model's usefulness for making predictions. In calculating Q^2 , we validate the endogenous components' predictive ability. If Q^2 is

greater than zero, this means that our reconstructed values are accurate and that the model is useful for making predictions (Geisser, 1974). The Q^2 values for this investigation are greater than 0 (as shown in Table 4), indicating that the model has a high predictive value.

The “finite mixture PLS module” was used to evaluate unobserved heterogeneity. The results for the entropy statistic were consistently substantially above 0.7 for a variety of different numbers of segments. Following the findings for the Bayesian and Akaike information criteria, the two-segment solution was shown to have the maximum positive support. Based on these findings, it seems that unobserved heterogeneity does not play a significant role here. In PLS-SEM, an overall measure of how well a model fits the data has been established and is called the goodness of fit. When researchers are examining the PLS-SEM findings of various data groups using the same PLS path model, Henseler and Sarstedt (2013) demonstrate that the goodness of fit may be beneficial for a PLS multigroup analysis. The standardised root mean square residual (SRMR) is a goodness of fit metric for PLS-SEM that was introduced by Henseler et al. (2014). This measure may be used to avoid model misspecification. Hu and Bentler (1999) indicate that an optimal SRMR value is less than 0.08 or 0.10. The SRMR value for the present research is 0.039, which is less than 0.08, indicating a good model fit.

4.4. Hypothesis assessment

To evaluate the assumptions presented in this study, we bootstrapped the 142 samples received via the SmartPLS software, using a total of 5000 subsamples. Following this step, we determined whether the hypotheses were supported based on the path coefficients that were produced and their respective degrees of significance (Fig. 3). We employed a variety of standardised values as outputs to evaluate the direct influence that many independent factors have on many dependent variables. As a result of the fact that the β value for each of the hypotheses is positive, with corresponding p values of lower than 0.05, we may conclude that all eight hypotheses can be accepted. These findings are presented in Fig. 3 and Table 6.

5. Discussion

This research aimed to investigate the relationships between several firm characteristics, such as information and communication technology capabilities, firm technological readiness, and firm fit, as well as AIPRM capability, using a total of eight hypotheses. In addition, within the

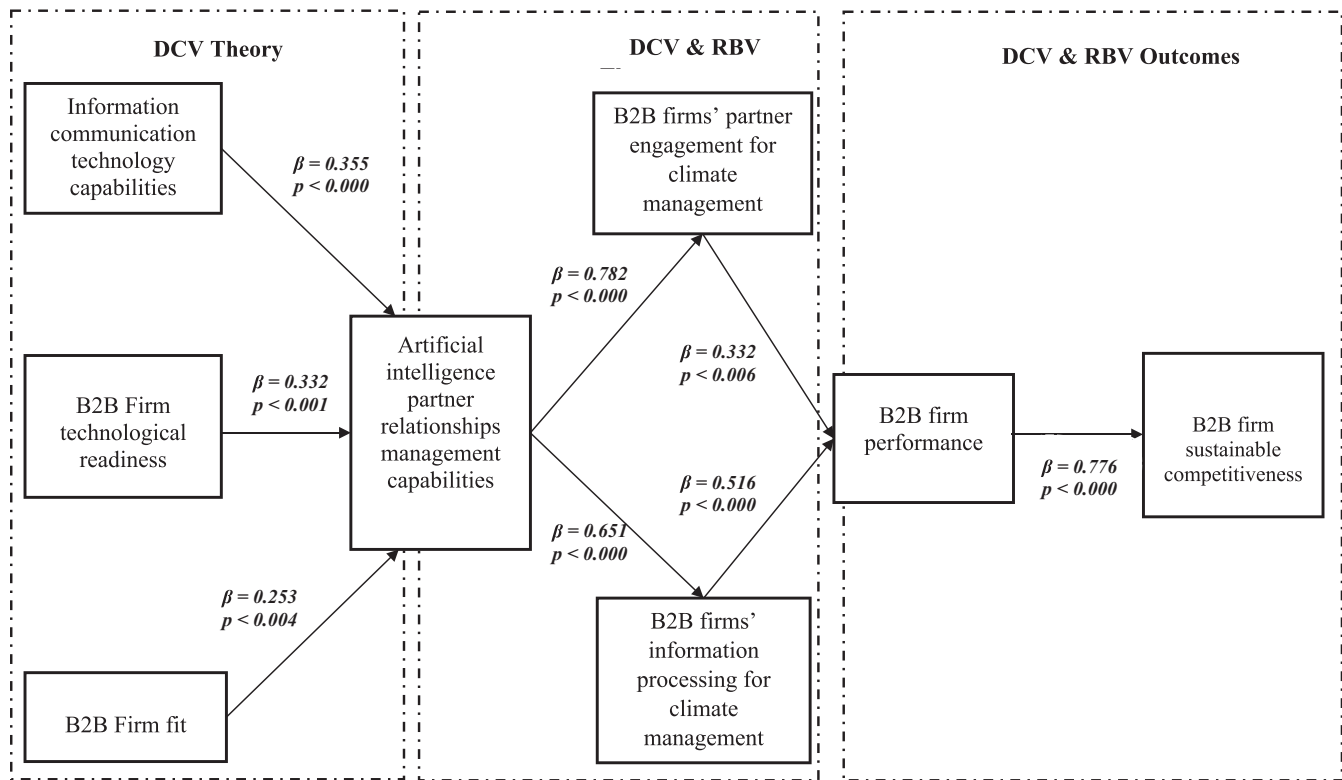


Fig. 3. Model with statistical results.

Table 6
Summary of the hypothesis testing results.

Path relation	β	SD	t-statistic	p-value	Hypothesis	Decision
ICTC → AIPRM	0.355	0.091	3.887	0.000	H1	✓
FTR → AIPRM	0.332	0.103	3.211	0.001	H2	✓
FF → AIPRM	0.253	0.089	2.861	0.004	H3	✓
AIPRM → FPE	0.782	0.066	11.893	0.000	H4	✓
AIPRM → FIP	0.651	0.077	8.422	0.000	H5	✓
FPE → FP	0.332	0.120	2.773	0.006	H6	✓
FIP → FP	0.516	0.120	4.299	0.000	H7	✓
FP → SC	0.776	0.058	13.272	0.000	H8	✓

✓ = Accepted.

context of these eight hypotheses, we have investigated the relationships between AIPRM, firm partner engagement and firm information processing for climate management, as well as the paths these relationships take to achieve firm performance and sustainable competitiveness. The PLS-SEM methodology was used to conduct statistical analysis and verify the assumptions. It has been noted that each of the hypotheses was shown to be supported.

According to the findings of the research, among the effects of information and communication technology capabilities on AIPRM (H1), firm technological readiness on AIPRM (H2), and firm fit on AIPRM (H3), the effect of information and communication technology capabilities on AIPRM (H1) is the most significant, given that the relevant path coefficient is 0.355 and the level of significance is $p < 0.000$. In addition, the implications of DCV theory regarding the effect of these three firm variables on AIPRM have been supported, as was previously noted. The validation of H1 is in line with the research of Chatterjee, Rana, et al. (2021). Similarly, H2 is supported by Chatterjee (2019) and Chatterjee, Ghosh, et al. (2021), and H3 is supported by Chatterjee, Rana, et al. (2021), Eriksson (2014), and Morgan, Vorhies, and Mason

(2009).

In addition, when seen through the prism of DCV and RBV theory, between the impacts of AIPRM on firm partner engagement for climate management (H4) and AIPRM on firm information processing for climate management (H5), the relationship between AIPRM and firm partner engagement for climate management (H4) is the strongest, with the concerned path coefficient being 0.782 with the level of significance at $p < 0.000$. Previous research theoretically supports H4 (Chatterjee, Rana, et al., 2021; Bag et al., 2021) and H5 (Lacka et al., 2020). Furthermore, when seen through the prism of DCV and RBV theory, when comparing the relationships between firm partner engagement for climate management and firm performance (H6) and firm information processing for climate management and firm performance (H7), the relationship between firm information processing for climate management and firm performance (H7) is the strongest, since the relevant path coefficient is 0.516, at a significance level of $p < 0.000$. Some previous studies also back up hypotheses H6 (Cainelli et al., 2015) and H7 (Farrokhi et al., 2020). The final eighth hypothesis, which proposes that there is a positive relationship between firm performance and sustainable competitiveness, is similarly supported by the statistical path coefficient value of 0.776 with a significance level of $p < 0.000$. Rahman et al. (2023) and Chatterjee, Rana, et al. (2021) also support that firm performance improves firm sustainable competitiveness.

6. Implications

6.1. Theoretical implications

To fully exploit AIPRM for environmentally efficient practices, businesses must invest significant resources into establishing and improving their information and communication technology capabilities (Rahman et al., 2023) and firm technological readiness capacities (Telukdarie et al., 2018). This necessitates persistent spending on research and development, infrastructure upgrades, and staff education and development (Gupta and George, 2016). Firm fit also suggests that

the company focus its attention on building assets and skills that are in step with its strategic objectives and external environment (Chatterjee, Rana, et al., 2021). Companies must have a teachable stance regarding information and communication technology capabilities, firm technological readiness, and AI technology. As part of this process, they should keep an eye on developments in the industry and new technology, testing out novel applications, adapting to a shifting climate, and soliciting constructive criticism from business partners and consumers (Chatterjee, Ghosh, et al., 2021). Firm fit recommends that the company take note of its past mistakes and make strategic changes as a result.

To effectively engage partners and enhance information processing systems, businesses must devote appropriate resources to creating and improving their AIPRM capabilities (Chatterjee, 2019). Long-term expenditures in research and development, infrastructure, and staff training and development are necessary for this to happen. Companies should view AIPRM and AI technology with a growth mindset. This requires keeping a close eye on developments in the industry and technology in general, trying out novel AIPRM and AI-based solutions, and assertively soliciting input from partners and consumers to pinpoint problem areas (Chatterjee, Ghosh, et al., 2021).

The RBV recognises the value of AIPRM as a tool for enhancing partner engagement and enhancing the efficiency of the company's data processing infrastructure while focusing on environmental strategies (Chatterjee, Ghosh, et al., 2021). Being uncommon, precious, difficult to replicate, and difficult to substitute, this resource can give a persistent competitive advantage. From the data-driven viewpoint, AIPRM is a capacity that allows the company to evolve and adapt to new circumstances as they arise in the marketplace, among other factors (Makadok, 2001). The company may obtain an advantage over the competition by investing heavily in the development of its AIPRM capabilities. The RBV and DCV perspectives, which stress the significance of developing and enhancing the firm's resources and capabilities to adapt and change in response to shifting market conditions, shifting customer needs, and emerging technological advancements (Aker et al., 2016), can be used to maximise the beneficial impact of AIPRM on firm partner engagement and firm information processing for managing climate in B2B firms. AIPRM and AI technologies can help the company improve partner engagement and information processing systems, giving them a competitive edge in the long term (Bag et al., 2021). This will be possible if the company takes a learning-oriented approach, forms collaborative partnerships, and adopts a flexible and agile organisational structure.

In the context of firm partner engagement, this means that organisations should cultivate and utilise skills and assets that boost partner engagement, such as strong communication, teamwork, and relationship management while focusing on strategies based on climate actions. To boost trust, loyalty, and customer satisfaction, businesses may use these tools and resources to create solid bonds with their business associates (Pillai, Sivathanu, & Dwivedi, 2020). Improved performance results can follow, including more business and money made per unit of effort (Dwivedi, Ismagilova, Rana, & Raman, 2021). Additionally, this implies that businesses should cultivate flexible capabilities that allow them to respond to partner needs and preferences, including the ability to tailor products and services, create novel offerings, and provide individualised customer service, while maintaining a sharp focus on climate-based actions (Chatterjee, Rana, et al., 2021). With these skills, businesses may create long-lasting bonds with their partners, which can boost their performance in a variety of ways.

RBV and DCV can magnify the beneficial impacts of firm partner engagement and firm information processing on firm performance in a climate-focused context. Businesses may gain an edge in the market and improve their performance by investing in the growth of resources and skills that boost partner interaction and information processing for climate management in B2B firms. To further ensure their long-term success and competitiveness, businesses should cultivate dynamic skills that allow them to adapt and evolve in response to shifting market conditions and technological developments (Rahman et al., 2023).

Companies currently need to have a structure that can swiftly adjust to new technologies, consumer demands, and market shifts. To do this, a culture of innovation, experimentation, and risk-taking is necessary, as is the capacity for quick deployment and scaling of information and communication technology, firm technological readiness, and AI solutions (Chatterjee, Ghosh, et al., 2021). Firm fit recommends that the company modify its structure to better accommodate its strategic objectives and the market.

6.2. Managerial implications

To gain access to the most recent innovations and develop climate-related best practices in information and communication technology, firm technological readiness, and AI, businesses must collaborate with AI technology suppliers, industry experts, and academic institutions. For instance, the recent announcement of an extended partnership between Paige, a prominent provider of comprehensive digital pathology solutions and clinical AI applications, and Leica Biosystems, a leading company in the field of cancer diagnostics and workflow solutions (Businesswire, 2023). This collaboration aims to improve the implementation of digital pathology workflows in medical facilities and laboratories worldwide. Paige anticipates offering a comprehensive end-to-end solution that facilitates the adoption of digital pathology methods for medical facilities worldwide, leveraging the extensive coverage of Leica Biosystems scanners (Businesswire, 2023). To better understand their wants, requirements, and expectations, this requires the establishment of a solid rapport with partners and consumers. Firm fit suggests that the company seeks out alliances that will help it strengthen its weak spots.

Overall, DCV may capitalise on the positive impact of information and communication technology, firm technological readiness, and firm fit on AIPRM by stressing the significance of strengthening the company's ability to change and adapt in response to shifting market conditions, consumer preferences, and technical developments (Makadok, 2001). A company may successfully employ AIPRM and obtain a competitive edge in the market if it adopts a learning orientation, builds collaborative partnerships, and adopts an agile and adaptable organisational structure (Eriksson, 2014). In addition to improving the company's AIPRM capacities, the model provided by firm fit may help it better align its resources and capabilities with its strategic goals and the competitive environment.

As the effects of climate change become more severe, businesses will need to implement new strategies in response to shifting market conditions, shifting customer desires, and emerging technologies. The capacity to swiftly deploy and expand AIPRM and AI solutions, as well as a culture that encourages innovation, risk-taking, and experimentation, is necessary. For instance, according to Sebastian Steinhäuser, the Chief Strategy Officer at SAP SE, their collaboration with Icertis serves as a valuable addition to the SAP portfolio. By incorporating contract intelligence as a crucial system of recording throughout the SAP suite, they can seamlessly integrate potent contract insights into core business processes. Furthermore, Steinhäuser also stated that the progress observed with Icertis in the past year is merely the initial phase, and there is a desire to extend the collaboration to facilitate enterprises in realising the full potential of their contracts throughout the organisation (Haas & Wright, 2023).

The financial institution, Deutsche Bank, has recently disclosed a collaborative innovation partnership with NVIDIA spanning multiple years, intending to expedite the integration of AI and machine learning technologies within the financial services industry (Deutsche Bank, 2022). The amalgamation of Deutsche Bank's proficiency in the financial industry with NVIDIA's dominance in AI and accelerated computing will expedite the advancement of a diverse array of AI-driven services that comply with regulatory standards (Deutsche Bank, 2022).

In the context of firm information processing, RBV proposes that businesses create and use data analytics, information management

systems, and knowledge-sharing platforms as efficient information processing methods for developing better actions against climate change (Farrokhi et al., 2020). Furthermore, better performance outcomes, such as greater operational efficiency, innovation, and competitive advantage, may be achieved using these resources and skills by extracting important insights from data and enhancing decision-making processes. For instance, [Leinwand and Mani \(2021\)](#) present an analysis covering many companies that have implemented substantial changes to their operations to adapt to the digital era.

7. Conclusion

This study investigates the external factors that affect the effective adoption of AIPRM skills and how such capabilities influence firm performance and businesses' ability to maintain a competitive advantage over the long term when the firm's intended activities are climate-based. This research has shown that dynamic capabilities are crucial to enabling businesses to embrace and effectively utilise AIPRM capabilities in the context of climate measures for competitive advantage, and this has been demonstrated through an examination of information and communication technology, firm technological readiness, firm fit, firm partner engagement for climate management, and firm information processing for climate management. The research has also identified AIPRM as a critical resource that can help cultivate the nimble expertise needed to deliver firm partner engagement and firm information processing effectively for managing climate in B2B firms. The study has also investigated the interrelationships between these factors, which are classified according to the RBV and DCV, as well as how they might improve firm performance and lead to enduring competitive advantage. This research has made important contributions to understanding the function of AIPRM in B2B firms by presenting a statistically oriented discussion, consequences, and future proposals based on the findings.

The outcomes of this research highlight the relevance of AIPRM as a strategic tool that may assist B2B firms in reacting to a commercial environment that is becoming increasingly dynamic and competitive. The necessity of companies adopting a dynamic capabilities viewpoint, which allows them to adapt and react to changing market conditions successfully, is underlined here. This helps companies adapt and respond to changing market situations effectively. The research also highlights the potential for AIPRM to ease firms' adoption of environmentally friendly practices and bestow a competitive advantage in the marketplace. This study offers significant insights that may aid businesses in utilising the full potential of AIPRM to encourage development and prosperity despite the obstacles introduced by an ever-evolving business environment. These insights can be used to guide organisations in exploiting AIPRM.

7.1. Limitations and future research directions

In terms of potential limitations, we may acknowledge that the current research focuses only on the external elements that influence adoption and implementation; no consideration is given here to the internal aspects that may affect a company's capacity to build AIPRM capabilities. Additionally, this research does not consider possible barriers to adopting and implementing AIPRM capabilities, such as employee aversion to change, organisational culture, or costs associated with these factors. Because of the elapsed time between the period covered in this study and the present day, The potential advantages of deploying AIPRM capabilities are given precedence in this research above any possible restrictions or constraints that may occur in practice.

Furthermore, based on the findings of this study, we realised that they are still much to do in this emerging area of research. Therefore, we have proposed many future research questions, which researchers may wish to explore in the context of recent and emerging research (as shown in [Table 7](#)).

Table 7
Proposed future research questions based on the study findings.

Associated paths	Proposed future research questions (PFRQ)
ICTC → AIPRM (H1) FTR → AIPRM (H2) FF → AIPRM (H3)	PFRQ1: How can B2B firms utilise their information and communication technology capabilities to boost AIPRM in the context of climate change practices, especially in lowering greenhouse gas emissions, encouraging sustainability, and partnering on green projects? PFRQ2: How can several components of information and communication technology capabilities, such as technological infrastructure, digital abilities, and innovation features, impact AIPRM in tackling climate change practices in B2B firms? PFRQ3: What are the constraints and facilitators of utilising information and communication technology capabilities to increase AIPRM in the setting of global warming behaviour, and how can B2B firms address these obstacles? PFRQ4: How can AIPRM and information and communication technology capabilities collaborate better with corporate frameworks and policies to encourage environmental mitigation and sustainability in B2B collaborations? PFRQ5: To what extent does environmental uncertainty moderate the relationships between information and communication technology capabilities, firm technological readiness, firm fit, and adoption of AIPRM practices for sustainable development? PFRQ6: How can B2B firms employ their information and communication technology capabilities, firm technological readiness, and firms fit to increase AIPRM in the framework of green financing, sustainable supply chain management, and various other climate change adaptations and mitigation efforts? PFRQ7: What are the key challenges and opportunities that B2B firms face in adopting AIPRM practices for achieving net-zero emissions, and how can their information and communication technology capabilities, firm technological readiness, and firm fit be leveraged to overcome these challenges and seize these opportunities? PFRQ8: What are the implications of adopting AIPRM practices for sustainable development on the competitiveness and long-term success of B2B firms, and how can information and communication technology capabilities, firm technological readiness, and firm fit contribute to this success? PFRQ9: How can B2B firms measure and evaluate the impact of AIPRM practices on their sustainability goals, and what role do information and communication technology capabilities, firm technological readiness, and firm fit play in this process? PFRQ10: What are the ethical and social implications of adopting AIPRM practices for sustainable development, and how can B2B firms ensure that their practices align with the values and expectations of their stakeholders? PFRQ11: How does AIPRM affect firm partner engagement in the context of net-zero emissions, and how can B2B firms use AIPRM to improve their engagement with partners in achieving sustainability goals? PFRQ12: What is the impact of AIPRM on firm information processing in the context of green financing, and how can B2B firms use AIPRM to improve their information processing for sustainable financial decision-making?
	PFRQ13: How might AIPRM impact firm partner engagement and firm information processing in the climate change action framework, and what are the primary operations driving these consequences?
	PFRQ14: What are the key factors that affect the adoption and success of AIPRM in the metaverse, particularly concerning firm partner engagement and firm information processing, and how can B2B firms overcome these challenges?
	PFRQ15: What is the relationship between AIPRM, firm partner engagement and firm information processing, and how can these variables be leveraged to achieve sustainable development goals, such as net-zero emissions, green financing, and Industry 5.0?
	PFRQ16: How can B2B firms utilise AIPRM to interact with collaborators and wider society on environmental problems, and what are the advantages and obstacles involved with such endeavours?
	PFRQ17: How can B2B firms address the ethical and social implications of AIPRM in their partner engagement and
AIPRM → FPE (H4)	
AIPRM → FIP (H5)	

(continued on next page)

Table 7 (continued)

Associated paths	Proposed future research questions (PFRQ)
	information processing, particularly in the context of achieving sustainability goals? PFRQ18: How can AIPRM be integrated with other emerging technologies, such as blockchain and the Internet of Things, to further enhance firm partner engagement and firm information processing for sustainable development? PFRQ19: What are the potential barriers and challenges that B2B firms may face in adopting AIPRM for achieving sustainability goals, and how can these challenges be overcome? PFRQ20: What are the implications of AIPRM for the competitive advantage of B2B firms, and how can they use AIPRM to differentiate themselves in the market and achieve long-term success? PFRQ21: How does firm partner engagement impact firm performance for firm sustainable competitiveness in the context of net-zero emissions, and what role does firm information processing play in this relationship? PFRQ22: How does firm information processing affect firm performance for firm sustainable competitiveness in the context of green financing, and how does firm partner engagement mediate this relationship? PFRQ23: What is the relationship between firm partner engagement, firm information processing, and firm performance for sustainable competitiveness in the context of Industry 5.0, and how can B2B firms use this relationship to gain a competitive advantage? PFRQ24: How does adopting emerging technologies, such as AIPRM, impact the relationship between firm partner engagement, firm information processing, and firm performance for sustainable competitiveness in the metaverse? PFRQ25: What are the key factors that affect the relationship between firm partner engagement, firm information processing, and firm performance for sustainable competitiveness, and how can B2B firms leverage these factors to achieve SC in the long term? PFRQ26: How can B2B firms measure and evaluate the impact of firm partner engagement, firm information processing, and firm performance for sustainable competitiveness, and what key performance indicators should be used in this evaluation? PFRQ27: How can B2B firms address the ethical and social implications of firm partner engagement, and firm information processing in pursuing sustainable competitiveness, particularly in the context of net-zero emissions, green financing, Industry 5.0, and the metaverse? PFRQ28: What are the potential barriers and challenges that B2B firms may face in improving their firm partner engagement and firm information processing for sustainable competitiveness, and how can these challenges be overcome? PFRQ29: How can B2B firms integrate sustainability considerations into their firm partner engagement and firm information processing strategies to achieve SC? PFRQ30: What are the implications of firm partner engagement and firm information processing for the long-term success of B2B firms, and how can they use these strategies to differentiate themselves in the market and achieve sustainable competitive advantage?
FPE → FP (H6)	
FIP → FP (H7)	
FP → SC (H8)	

Data availability

Data will be made available on request.

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